

CS102 – Algorithms and Programming II
Programming Assignment 3
Fall 2025-2026

ATTENTION:

- Compress all of the Java program source files (.java) files into a single zip file.
- The name of the zip file should follow the below convention, where you replace the variables “Sec1”, “Surname” and “Name” with your actual section, surname and name:
CS102_Sec1_Asst3_Surname_Name.zip
- Upload the above zip file to Moodle by the deadline (if not, significant points will be taken off). You will get a chance to update and improve your solution by consulting to the TAs and tutors during the lab. You have to make the preliminary submission before the lab day. After the TA checks your work in the lab, you will make your final submission. Even if your code does not change in between these two versions, you should make two submissions for each assignment.

GRADING WARNING:

- Please read the grading criteria provided on Moodle. The work must be done individually. Code sharing is strictly forbidden. We are using sophisticated tools to check the code similarities. The Honor Code specifies what you can and cannot do. Breaking the rules will result in disciplinary action.

CARGO DELIVERY SYSTEM

For this lab assignment, you will implement a Java console application simulating a delivery system with different staff and vehicles. You are expected to use object-oriented programming, namely class hierarchies (inheritance), to avoid code repetition and polymorphism. The simulation will be on a turn basis. At each turn, the user can purchase new vehicles, hire new staff members, and assign them to different vehicles to perform their tasks. The simulation aims to grow the delivery service while keeping a positive income.

Each vehicle has a cargo capacity, staff capacity, a unique plate number (randomly generated number when the user buys a specific vehicle), buying cost, and speed. Include a parent Vehicle class so that you can extend it with the following child classes:

Type	Cargo Capacity	Staff Capacity	Cost	Speed
Bicycle	5	1	100	10
Car	20	2	2000	50
Van	100	4	4000	25
Truck	200	5	7000	20
Helicopter	50	2	15000	100

Implement a **Staff** class to represent the staff members. Each staff member should have a name, surname, salary (this amount is deducted from the user’s total money each turn), and skills. The skills of the staff members are based on their types. You should have the following subclasses of the Staff:

- **Driver:** This class includes a driving speed limit (a random integer between 10 and 100) and a delivery capacity (a random integer between 3 and 10).
- **Carrier:** This class includes a price boost (a random integer between 0 and 5) and delivery capacity (a random integer between 5 and 25).
- **Receiver:** This class includes a receiving capacity (a random integer between 10 and 20).

Each staff member can be assigned to specific jobs. Drivers and carriers are manually assigned to the vehicles to function. Receivers are automatically assigned to the related facility. The user can have any number of staff members. There is no upper limit for the working receivers. However, a driver or carrier must be assigned to a vehicle to perform their tasks. The user can buy any number of vehicles. A vehicle should have at least one driver to work; it does not need to fill its staff capacity. A vehicle's number of staff members cannot exceed its staff capacity.

Whether assigned to a vehicle or not, the salary of all the staff members hired by the user is subtracted from the total money every turn. Unlike staff members, the cost of vehicles is a one-time payment, subtracted from the total money when the user buys the vehicle.

The vehicles are standard, and the user can buy any number of any vehicle at the beginning of the turn. However, the staff members are a limited resource. Five new staff members will be generated and available for hire at the beginning of each turn. The user will choose staff members from this limited set. The user may hire all five available staff members, or any number. Once all five available staff are hired, they are not refreshed during the same turn so that the user can hire at most five members each turn. You should generate a new set of staff members available for hire at each turn. For example, if the user does not hire a specific person, the same person will not be available for the next turn.

Any staff hired can be reassigned as many times as needed. Whether a staff member is assigned to a vehicle or not, their salary is still deducted from the total money, so it is more meaningful to keep them all assigned. The set of 5 available staff for hire is generated based on the following rules:

- The name and surname of the staff should be randomly selected from an array of names and surnames you make.
- The staff's type (driver, carrier, receiver) is determined randomly. The set of five can be any combination; for example, two drivers and three receivers could be available for hire in a certain turn.
- The skills of the staff are determined randomly based on the given ranges.
- The salary is calculated as follows for each staff type (you will use integer division):
 - Driver: $(\text{speed limit} / 10) + (\text{delivery capacity} / 2)$,
 - Carrier: $\text{price boost} + (\text{delivery capacity} / 5)$, and
 - Receiver: $\text{receiving capacity} / 2$.

The staff with higher skills requires a higher salary. The user will try to find a balance to maximize profit. The company earns money from delivered packages. A package should first be received in order to be delivered. After the user makes the decisions for that turn, choosing the “end turn” option triggers the following steps:

- The sum of the receiving capacity of all receiver staff is added to the total number of company packages.
- For each vehicle with at least one driver, the following stats are calculated:

- The effective speed of the vehicle is calculated as the minimum of the vehicle's speed and the driver's speed limit. If no driver is assigned to the vehicle, the effective speed is 0. If multiple drivers are assigned, a higher speed limit is used in the calculation above.
- The effective delivery capacity is calculated as the sum of the delivery capacities of all drivers and carriers in the vehicle. However, this number cannot exceed the vehicle's cargo capacity.
- The current delivery count is the minimum of the company's total number of packages and the effective delivery capacity of this vehicle. Once the current delivery count is calculated, subtract it from the total packages. Each vehicle is getting its packages from the total count; a vehicle cannot deliver if a certain package has not been received yet. The packages are given to the vehicles on a first-come, first-served basis. If there are not enough received packages, certain vehicles can be empty.
- The effective price boost is calculated as the sum of the price boosts of all the carriers in the vehicle. However, this number cannot be less than 1. If no carriers are in the vehicle, the effective price boost equals 1.
- The profit made by this vehicle is calculated as follows:

$$\text{effective price boost} \times \text{effective delivery capacity} \times (\text{effective speed} / 10).$$
- The sum of the profit of all vehicles with at least one staff member in them is added to the total money.
- The sum of the salaries of all staff members is subtracted from the total money.
- The salary of a randomly selected staff member is increased by 5.

At the beginning of the turn, the user has the following options:

- **Buy a vehicle:** The vehicles are listed with their stats and cost. The user can choose and buy any number of vehicles.
- **List owned vehicles:** The vehicles are listed with their type and unique plate number. For each vehicle, the staff assigned to the vehicle is also listed, as well as the current profit of that vehicle.
- **Sell a vehicle:** The user can choose a vehicle to sell. Half of the original cost is added back into the total money. Any staff assigned to the vehicle is automatically unassigned.
- **Hire staff:** List the set of 5 randomly generated staff for hire, and let the user hire the chosen ones. The salary of the newly hired staff members is deducted at the end of the turn, together with the other staff, so there is no immediate payment while hiring.
- **Assign staff:** Let the user assign any unoccupied staff member to a vehicle. This option is valid for carriers and drivers; receivers do not need to be assigned. This option should list the staff members (excluding the receivers) with their currently assigned vehicles. Let the user choose a staff member to assign using the order numbers of the vehicles. Then, list the vehicles with available staff capacity so the user can choose a vehicle to assign the selected staff member. The user can also choose none to unassign the staff member.
- **Distribute staff:** This option distributes unassigned staff members to available vehicles randomly. It will be used as an easy way of filling the vehicles. Make sure this option first assigns a driver to the empty vehicles so that they can function.
- **Fire staff:** List the staff members' currently assigned vehicles and salaries. Let the user choose a staff member by their order number to fire. Any staff member fired is removed from the staff list, and their salary is no longer subtracted from the total money.

- **End turn:** Ends the current turn to begin the next one. Once this option is selected, the “end turn” steps are performed, and the user's total money is updated.

The user starts with 500 money units.

Preliminary Submission: You will submit an early version of your solution before the final submission. This version should at least include the following:

- The basic structure of the vehicle and staff classes, with their subclasses, should be completed.
- We should be able to hire staff members and assign them to existing vehicles.

Not completing the preliminary submission on time results in 50% reduction of this assignment's final grade.